# Supplemental material - week 12.

Thorough discussion of Example 11.5 from the clustering lecture. Step-by step distance matrices for

## i) Single Linkage

$D_1$ :

| | 1 | 2 | {3,5} | 4 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | 50 | 73 | 98 | 41 | 65 |
| 2 | | 0 | 20 | 41 | 80 | 37 | 65 |
| {3,5} | | | 0 | 2 | 25 | 18 | 34 |
| 4 | | | | 0 | 17 | 20 | 32 |
| 6 | | | | | 0 | 13 | 9 |
| 7 | | | | | | 0 | 4 |
| 8 | | | | | | | 0 |

$\Rightarrow$

$D_2$ :

| | 1 | 2 | {3,4,5} | 6 | 7 | 8 |
|---|---|---|---|---|---|---|
| 1 | 0 | 10 | 50 | 98 | 41 | 65 |
| 2 | | 0 | 20 | 80 | 37 | 65 |
| {3,4,5} | | | 0 | 17 | 18 | 32 |
| 6 | | | | 0 | 13 | 9 |
| 7 | | | | | 0 | 4 |
| 8 | | | | | | 0 |

$\Rightarrow$

$D_3$ :

| | 1 | 2 | {3,4,5} | 6 | {7,8} |
|---|---|---|---|---|---|
| 1 | 0 | 10 | 50 | 98 | 41 |
| 2 | | 0 | 20 | 80 | 37 |
| {3,4,5} | | | 0 | 17 | 18 |
| 6 | | | | 0 | 9 |
| {7,8} | | | | | 0 |

$D_4$ :

| | 1 | 2 | {3,4,5} | {6,7,8} |
|---|---|---|---|---|
| 1 | 0 | 10 | 50 | 41 |
| 2 | | 0 | 20 | 37 |
| {3,4,5} | | | 0 | 17 |
| {6,7,8} | | | | 0 |

$\Downarrow$

And we end up with: $\{1,2\}$ $\{3,4,5\}$ $\{6,7,8\}$ as the 3 clusters.

On the next page I describe the cluster agglomeration by using the complete linkage approach. We will have different agglomerations in the process but we end up with the same 3 clusters at the end:

ii) Complete linkage

$\tilde{D}_1$:

| | 1 | 2 | {3,5} | 4 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | 53 | 73 | 98 | 41 | 65 |
| 2 | | 0 | 25 | 41 | 80 | 37 | 65 |
| {3,5} | | | 0 | 5 | 25 | 18 | 34 |
| 4 | | | | 0 | 17 | 20 | 32 |
| 6 | | | | | 0 | 13 | 9 |
| 7 | | | | | | 0 | 4 |
| 8 | | | | | | | 0 |

$\tilde{D}_2$:

| | 1 | 2 | {3,5} | 4 | 6 | {7,8} |
|---|---|---|---|---|---|---|
| 1 | 0 | 10 | 53 | 73 | 98 | 65 |
| 2 | | 0 | 25 | 41 | 80 | 65 |
| {3,5} | | | 0 | 5 | 25 | 34 |
| 4 | | | | 0 | 17 | 32 |
| 6 | | | | | 0 | 13 |
| {7,8} | | | | | | 0 |

$\tilde{D}_3$:

| | 1 | 2 | {3,4,5} | 6 | {7,8} |
|---|---|---|---|---|---|
| 1 | 0 | 10 | 73 | 98 | 65 |
| 2 | | 0 | 41 | 80 | 65 |
| {3,4,5} | | | 0 | 25 | 34 |
| 6 | | | | 0 | 13 |
| {7,8} | | | | | 0 |

$\tilde{D}_4$:

| {1,2} | {3,4,5} | 6 | {7,8} |
|---|---|---|---|
| 0 | 73 | 98 | 65 |
| | 0 | 25 | 34 |
| | | 0 | 13 |
| | | | 0 |

$\Rightarrow$ So we end up with {1 2}, {3,4,5}, {6,7,8} again